Macadamia Fertilizer Programme 2026/27.

Fertilise Smarter, Grow Stronger.

Season 2026/27 programme for a 5.00 ha block of macadamia.

PREPARED FOR	Demo / validation
PREPARED BY	Sapling Fertilizer
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REFERENCE BLOCK — MACADAMIA · LEVUBU REFERENCE · LEVUBU, LIMPOPO

SAPLING · AGRICULTURAL ADVISORY

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SECTION 01

Background.

This programme covers the 2026/27 season for macadamia across 1 block totalling 5.00 ha. The strategy is built around the soil analysis, the available equipment, and the crop's stage-by-stage demand curve.

PROGRAMME AT A GLANCE

- 1 block totalling 5.00 ha
- 10 applications across 5 blends this season
- 3 targeted foliar sprays addressing soil-availability gaps and stage-peak demand
- 3 open items flagged for tracking through the season

SECTION 02

Soil Report.

Soil analysis for Reference Block A4 (Levubu) (5.00 ha) was reviewed against the crop's nutrient requirements for the season.

Reference Block A4 (Levubu) 5.00 ha

Reference block (published norms) · Mehlich-3 · sampled 1 Mar 2026

Ratios & balance checks

RATIO	READING	TARGET	STATUS	POSITION VS IDEAL
Ca : B	2,800 : 1	≤ 1000 : 1	WARN	
	Soil Ca:B = 2800 exceeds 1000:1 — Ca-induced B deficiency likely, especially at flowering. *Apply foliar B at critical stages (pre-bloom, fruit set)*			
P : Zn	7.3 : 1	≤ 150 : 1	OK	
	Within target — no action.			
Sodium saturation (ESP)	1.0 %	≤ 5 %	OK	
	Within target — no action.			

The loudest signals

> Soil Ca:B = 2800 exceeds 1000:1 — Ca-induced B deficiency likely, especially at flowering.

> pH (KCl) 4.80 below crop-optimal band (5.0–6.5). Acidic soil restricts Ca / Mg / P availability; root expansion limited.

SECTION 03

Pre-Season Actions.

Corrective amendments that need to be on the ground before the in-season programme begins. Rates are per-block — each block gets the rate its soil chemistry calls for, even when grouped with other blocks for the in-season programme. Lime / gypsum / dolomite spreaders apply per-area, so per-block precision is operationally easy.

Group A

All blocks: Reference Block A4 (Levubu)

BLOCK	MATERIAL	RATE	APPLY BY	REACTION
Reference Block A4 (Levubu)	Calcitic Lime	1 500-2 500 kg/ha (size to lime requirement test)	Early-November 2026	~4.0 mo

Address: pH (KCl) 4.80 below crop-optimal band (5.0–6.5). Acidic soil restricts Ca / Mg / P availability; root expansion limited.

Cycle already underway — apply at the earliest opportunity to set up full reacted for the next cycle (target reaction window: ~4 months before 2027-03-01).

SECTION 04

Application Walkthrough.

Across the season you'll do 10 fertilizer applications across 1 block (5.0 ha total) of macadamia. Each one is laid out below in calendar order — what to apply, where, how, and why.

MARCH 2026	APRIL 2026	MAY 2026	JUNE 2026	JULY 2026
2 applications	2 applications	1 application	1 application	1 application

AUGUST 2026	SEPTEMBER 2026	OCTOBER 2026
1 application	1 application	1 application

A Group A

1 block · 5.0 ha · 10 applications

Reference Block A4 (Levubu)

Mid-March 2026

2 PARALLEL PASSES

Two parallel passes on the same day, each via the method that suits its nutrient profile. The two passes together give the trees what they need at this stage.

A1 PASS 1 OF 2 FERTIGATION

Inject 15:0:0 (16) + Ca 19.0% at 47 kg/ha

HOW: Inject through the drip irrigation line during a normal watering cycle.

TOTAL THIS PASS: 235 kg (5.0 ha × 47 kg/ha)

PER HA THIS PASS DELIVERS

N 14.5 kg | Ca 17.8 kg

WHY THIS PASS

Trees are at early-season re-growth — building roots, shoots, and the season's canopy, so this pass leads with calcium and nitrogen. Delivered through the irrigation system so it reaches the active root zone without an extra ground operation.

Mid-April 2026

2 PARALLEL PASSES

Two parallel passes on the same day, each via the method that suits its nutrient profile. The two passes together give the trees what they need at this stage.

A2 PASS 2 OF 2 DRY BROADCAST

Spread 6:5:15 (10) + Ca 2.0% + Mg 0.9% + S 0.9% at 276 kg/ha

HOW: Spread evenly under the canopy with a granular spreader.

TOTAL THIS PASS: 1 378 kg (5.0 ha × 276 kg/ha)

A3 PASS 1 OF 2 FERTIGATION

PER HA THIS PASS DELIVERS

Inject 15:0:0 (16) + Ca 19.0% at 47 kg/ha

A4 PASS 2 OF 2 DRY BROADCAST

A5

Mid-May 2026

Week 11 of season

FERTIGATION

Inject 8:4:15 (41) + Ca 7.6% + S 2.3% at 80 kg/ha

HOW	Inject through the drip irrigation line during a normal watering cycle.	TOTAL THIS PASS	399 kg of fertilizer (5.0 ha × 80 kg/ha)
STAGE	Vegetative — *leaf and shoot growth, building this year's canopy*		

WHY THIS ONE

Trees are at leaf and shoot growth, building this year's canopy, so this pass leads with potassium and nitrogen. Delivered through the irrigation system so it reaches the active root zone without an extra ground operation.

PER HECTARE THIS PASS DELIVERS

N 29.0 kg	P 15.0 kg	K 54.3 kg	Ca 18.1 kg	S 5.5 kg

A6

Mid-June 2026

Week 16 of season

FERTIGATION

Inject 8:4:15 (41) + Ca 7.6% + S 2.3% at 80 kg/ha

HOW	Inject through the drip irrigation line during a normal watering cycle.	TOTAL THIS PASS	399 kg of fertilizer (5.0 ha × 80 kg/ha)
STAGE	Vegetative — *leaf and shoot growth, building this year's canopy*		

WHY THIS ONE

Trees are at leaf and shoot growth, building this year's canopy, so this pass leads with potassium and nitrogen. Delivered through the irrigation system so it reaches the active root zone without an extra ground operation.

PER HECTARE THIS PASS DELIVERS

N 29.0 kg	P 15.0 kg	K 54.3 kg	Ca 18.1 kg	S 5.5 kg

A7

Mid-July 2026

Week 20 of season

FERTIGATION

Inject 8:4:15 (41) + Ca 7.6% + S 2.3% at 80 kg/ha

HOW	Inject through the drip irrigation line during a normal watering cycle.	TOTAL THIS PASS	399 kg of fertilizer (5.0 ha × 80 kg/ha)
STAGE	Vegetative — *leaf and shoot growth, building this year's canopy*		

WHY THIS ONE

Trees are at leaf and shoot growth, building this year's canopy, so this pass leads with potassium and nitrogen. Delivered through the irrigation system so it reaches the active root zone without an extra ground operation.

PER HECTARE THIS PASS DELIVERS

N 29.0 kg	P 15.0 kg	K 54.3 kg	Ca 18.1 kg	S 5.5 kg

A8

Mid-August 2026

Week 24 of season

FERTIGATION

Inject 5:2:15 (51) + Ca 3.4% + S 1.8% at 182 kg/ha

HOW	Inject through the drip irrigation line during a normal watering cycle.	TOTAL THIS PASS	910 kg of fertilizer (5.0 ha × 182 kg/ha)
STAGE	Reproductive — *flowering and fruit-set — pushing flower spikes and starting nuts*		

WHY THIS ONE

Trees are at flowering and fruit-set — pushing flower spikes and starting nuts, so this pass leads with potassium and nitrogen. Delivered through the irrigation system so it reaches the active root zone without an extra ground operation.

PER HECTARE THIS PASS DELIVERS

N 43.5 kg	P 18.0 kg	K 122.3 kg	Ca 12.5 kg	S 6.6 kg

A9

Mid-September 2026

Week 29 of season

FERTIGATION

Inject 5:2:15 (51) + Ca 3.4% + S 1.8% at 182 kg/ha

HOW	Inject through the drip irrigation line during a normal watering cycle.	TOTAL THIS PASS	910 kg of fertilizer (5.0 ha × 182 kg/ha)
STAGE	Reproductive — *flowering and fruit-set — pushing flower spikes and starting nuts*		

WHY THIS ONE

Trees are at flowering and fruit-set — pushing flower spikes and starting nuts, so this pass leads with potassium and nitrogen. Delivered through the irrigation system so it reaches the active root zone without an extra ground operation.

PER HECTARE THIS PASS DELIVERS

N 43.5 kg	P 18.0 kg	K 122.3 kg	Ca 12.5 kg	S 6.6 kg

A10

Mid-October 2026

Week 33 of season

FERTIGATION

Inject 6:2:15 (47) + Ca 4.9% + S 1.5% at 293 kg/ha

HOW	Inject through the drip irrigation line during a normal watering cycle.	TOTAL THIS PASS	1 465 kg of fertilizer (5.0 ha × 293 kg/ha)
STAGE	Fill — *nut / fruit fill — pulling nutrients into the kernel or fruit*		

WHY THIS ONE

Trees are at nut / fruit fill — pulling nutrients into the kernel or fruit, so this pass leads with potassium and nitrogen. Delivered through the irrigation system so it reaches the active root zone without an extra ground operation.

PER HECTARE THIS PASS DELIVERS

N 36.2 kg	P 12.0 kg	K 90.2 kg	Ca 14.2 kg	S 4.4 kg

SECTION 05

Foliar Sprays.

Targeted foliar applications addressing soil-availability gaps and stage-peak demand windows.

Group A

3 foliar events

WEEK	BLOCK	STAGE	ANALYSIS	RATE	REASON
6	Reference Block A4 (Levubu)	Vegetative I	20.5% B	1.5 kg	Soil Ca:B = 2800 exceeds 1000:1 — Ca-induced B deficiency likely, especially at flowering.
28	Reference Block A4 (Levubu)	Pre-flowering	20.5% B	0.1% spray	Critical for flower set + nut formation; B mobile in xylem only
32	Reference Block A4 (Levubu)	Vegetative flush	36% Zn	0.5% spray	Leaf expansion + enzyme activation; soil Zn often unavailable

SECTION 06

Season Totals.

Total applied this season per hectare, averaged across blocks. Drives shopping list and shed planning.

NUTRIENT	APPLIED PER HA
Nitrogen	~263 kg
Phosphorus	~114 kg
Potassium	~561 kg
Calcium	~151 kg
Magnesium	~10 kg
Sulphur	~44 kg

SECTION 07

Year 2 and Beyond.

Multi-year trajectory for this perennial programme.

THIS SEASON

Soil-side corrections in motion

- Pre-season lime / gypsum / dolomitic actions move soil chemistry in the right direction across Year 1, but most won't fully reflect in leaf or kernel until Year 2.
- In-season programme delivers the season's nutrient targets while the soil-side corrections work in the background.

YEAR 2 OUTLOOK

Where the soil should be

- Soil re-test mid-Year-2 to confirm pH / Ca / Mg movement from this season's amendments.
- Per-block rates revisited based on the re-test — blocks still trending wrong direction get a second pass; blocks that hit target shift to maintenance rates.
- Leaf analysis at flowering becomes the primary feedback loop once soil corrections are settled.

YEAR 3+

Maintenance trajectory

- Soil corrections largely complete — annual maintenance lime / gypsum + leaf-driven foliar plan.
- Programme stabilises around the orchard's productive rhythm; biggest variations come from yield carry-over and seasonal weather rather than soil-chemistry shifts.

SECTION 08

Items We're Carrying Into The Season.

Open questions, watchpoints, and defaults — grouped by when they need attention. Replaces the old split between "risk flags", "open items", and "assumptions"; one severity-banded list with cross-bucket duplicates removed.

Act before the season

These need attention up front — either too late to fully resolve this cycle, or material risks the engine wants flagged.

1. **Lime application — too late this season for full reaction.** Soil acidity / Al saturation needs lime to correct. Lime requires 3-6 months to react fully. Build-to-plant gap is only -1.8 months — insufficient for this season. Schedule lime for post-harvest of this crop to set up the next season. *Acidity / Al stress persists into the following season; early-growth and P availability will under-perform*

Watch through the season

Monitor these as the season progresses. Catch them on the next leaf or soil test, or on an in-season check-in.

1. **Irrigation water test (EC, pH, Ca, Mg, Na, HCO_3, Cl).** Required before first Part A injection. Dolomite-country water can produce scale when calcium-stream stock is injected. Borehole water EC + bicarb level affect drip-line pH. *Risk of emitter blockage from $CaCO_3$ precipitate*
2. **Mid-season leaf tissue analysis at tillering and at bulb-init/flowering.** Two natural decision points where the programme can still be steered. Catches deficiencies the soil test missed and validates actual uptake against plan. *Mid-season corrections become reactive, not anticipatory*

END OF PROGRAMME

Ref · REF-MAC-2026